MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FOURTH SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION

APRIL 2026

Fourth Semester

Major Course – MG4DSESDV200 – ELECTRO ACOUSTICS

(2023 Admission Onwards)

Duration: 2 Hours **Maximum Marks: 70**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I), Appreciation (Ap)

Students should attempt at least one question from each Course Outcome to enhance overall outcome attainability.

Part A **[Learning Domain] [CO No(s)]**

Very Short Answer Questions

Answer any **eight** questions. Each question carries **2 marks**. $(8 \times 2 = 16)$

1. Define electric current and state its SI unit. [K] [CO1]
2. Differentiate between AC and DC supply (any two points). [U] [CO1]
3. Name any two types of audio cables used in professional audio systems. [K] [CO2]
4. State one difference between balanced and unbalanced audio cables. [U] [CO2]
5. Define magnetic induction in the context of microphones. [K] [CO3]
6. What is the sensitivity of a microphone? [K] [CO3]
7. Name any two special-purpose microphones. [K] [CO3]
8. Name any two types of loudspeaker enclosures. [K] [CO4]
9. What is analog recording? [K] [CO4]
10. Name any two mass storage devices used in digital audio recording. [K] [CO4]

Part B

Short Answer Questions

Answer any **five** questions. Each question carries **6 marks**. $(5 \times 6 = 30)$

11. Describe the working principle of a capacitor and list its applications. [U] [CO1]
12. Describe the structure and use of a TRS cable in audio equipment. [U] [CO2]
13. Explain the directional characteristics of microphones and list any three types. [U] [CO3]
14. Describe the construction and working principle of a dual-element microphone. [U] [CO3]
15. Explain the following microphone characteristics: (a) Frequency Response (b) Signal-to-Noise Ratio [U] [CO3]
16. Explain the working principle of a moving coil loudspeaker with a neat diagram. [U] [CO4]
17. Describe the role of baffles and enclosures in loudspeaker performance. [U] [CO4]

Part C

Essay Questions

Answer any **two** questions. Each question carries **12 marks**. $(2 \times 12 = 24)$

18. Explain in detail the types, construction, working, and applications of different audio cables used in professional audio systems (XLR, TRS, TS, RCA) with neat diagrams. [An] [CO2]

19. Explain any five special-purpose microphones with construction, working, and applications: Lavalier, Shotgun, Boundary, Contact, and Binaural Microphone Systems. [An] [CO3]

20. Explain in detail the construction, working, and characteristics of a moving coil (electrodynamic) loudspeaker, including impedance, sensitivity, frequency response, power handling, and directivity. [An] [CO4]

END OF QUESTION PAPER

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